Data, information and units of storage

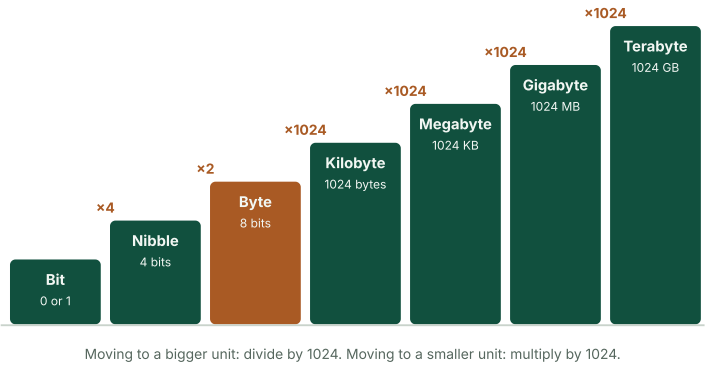
L01 FACT FILE

Specification: students should be able to

- Describe the difference between information and data
- Describe how data is stored in bit, nibble, byte, kilobyte, megabyte, gigabyte and terabyte
- Explain how characters are represented in binary using a character set such as ASCII
- Calculate conversions between units of storage

CCEA fact file: Digital Data 01: Representing Data

Key words: data, information, context, information system, binary, bit, nibble, byte, character set, ASCII, kilobyte, megabyte, gigabyte, terabyte, storage capacity



Moving to a bigger unit: divide by 1024. Moving to a smaller unit: multiply by 1024.

Units of storage as a staircase from bit to terabyte, with the multiplier between each step

Data and information

Data is raw facts and figures that have no meaning on their own. The numbers 07, 21 and 14 are data: you cannot tell what they are about. Information is data that has been processed and given a context, so it now has meaning. 'Enniskillen Rugby Club U16 results: won by 7 points, lost by 21 points, won by 14 points' is information, because we now know what the numbers describe. A useful way to remember this is that information is data with meaning. The same data can become different information depending on its context. For example, 140526 might be a date of birth (14/05/26), a product number in a shop or a winning lottery ticket. An information system, which is hardware and software working together, accepts data as input, processes it and outputs information. A supermarket till scanning barcodes is collecting data; the weekly sales report printed for the manager is information.

Bits, nibbles and bytes

Computers store everything in digital form using binary, a number system that has only two digits, 0 and 1. This suits computers because their circuits are built from switches that are either off (0) or on (1). One binary digit is called a bit, and it is the smallest unit of storage. Four bits make a nibble, which is half a byte and can store one hexadecimal digit. Eight bits make a byte. One byte is enough to store a single character, such as a letter, a digit or a punctuation mark. So the town name 'Strabane' has eight characters, takes eight bytes and needs 64 bits. Bits on their own can only represent two states, but grouping them allows many more values: a byte has 256 different patterns, from 00000000 to 11111111.

Character sets and ASCII

When you press a key, the computer needs a way to turn that character into binary. It does this using a character set, which is an agreed list that gives every character its own unique binary code. ASCII (American Standard Code for Information Interchange) is a widely used character set. In ASCII the capital letter A is 01000001, B is 01000010 and C is 01000011, so the codes for letters go up in order. Lower case letters have different codes: a is 01100001. Because both the sending and receiving computers use the same character set, text typed on one computer appears correctly on another. This is why a computer uses character codes: it can only store and process binary, and a shared code means text can be stored, sent and displayed reliably. Modern systems often use larger character sets, such as Unicode, to include symbols and other alphabets.

Units of storage and conversions

Storage capacity, such as the size of a USB flash drive, a hard drive or a computer's memory, is measured in bytes and multiples of bytes. Because computers count in binary, each unit is 1024 (which is 2 to the power of 10) times the one before. There are 1024 bytes in a kilobyte, 1024 kilobytes in a megabyte, 1024 megabytes in a gigabyte and 1024 gigabytes in a terabyte. To convert to a bigger unit, divide by 1024; to convert to a smaller unit, multiply by 1024. Remember that bits and bytes are linked by 8, not 1024. For example, 2 GB is 2 × 1024 = 2048 MB, and 12 288 KB is 12 288 ÷ 1024 = 12 MB. In 'how many will fit' questions, change both values into the same unit first, then divide the storage capacity by the size of one file. Check your answer makes sense before moving on.

T Key terms from the mark schemes

Data
Raw facts and figures that have no meaning or context on their own.
Credited: unprocessed facts and figures

Information
Data that has been processed and given context so that it has meaning.
Credited: data with meaning

Bit
A binary digit: the smallest unit of digital data, either a 0 or a 1 (off or on).
Credited: binary digit

Nibble
A group of 4 bits (half a byte). One hexadecimal digit can be stored in a nibble.

Byte
A group of 8 bits. One keyboard character (for example a letter, a digit or a punctuation mark) takes up 1 byte.
Credited: 8 binary digits

Kilobyte (KB)
1024 bytes – enough for a short plain-text file.

Megabyte (MB)
1024 kilobytes – around 1 minute of MP3 audio.

Gigabyte (GB)
1024 megabytes – around 20 minutes of HD video.

Terabyte (TB)
1024 gigabytes – the capacity of a typical laptop or external hard drive.

ASCII
American Standard Code for Information Interchange: a character set that gives each letter, digit and punctuation mark a unique binary code.
Credited: character set

Information system
Hardware and software working together to take data as input, process it and produce information as output.

Exam focus

This lesson appears in almost every Unit 1 paper. Data and information with examples was worth 4 marks in 2019 Q2(a): one mark for each definition and one for each suitable example, and the information example had to show context. 2023 Q1 asked about bits, bytes and data in context. Unit questions are often short: how many bits are in a nibble and how many kilobytes are in a megabyte (2019 Q2(b)), matching bit, nibble and byte to their definitions (2021 Q4), and converting 2 GB into megabytes (2025 Q4(c), answer 2048 MB). Common mistakes include writing 1000 instead of 1024, mixing up bits and bytes, dividing when you should multiply, and giving an 'information' example that is still just numbers. Use the exact terms bit, byte and binary digit, and always show your working in calculations, because method marks can be awarded even if the final answer is wrong.

Unit 1 2025 Q4	[7]	Boot process, OS roles, storage conversion
Unit 1 2023 Q1	[5]	Bit and byte, data types, data in context
Unit 1 2021 Q4	[6]	Bits, nibble, byte; data types matching
Unit 1 2019 Q2(b)	[2]	Bits in a nibble; KB in an MB
Unit 1 2019 Q2(a)	[4]	Data and information with examples

? Check yourself

1. Using an example of each, describe the difference between data and information. [4]
2. Put these units in order from smallest to largest: megabyte, bit, terabyte, byte, nibble, kilobyte, gigabyte. [2]
3. (a) How many bits are in a nibble? (b) How many kilobytes are in a megabyte? (c) How many bytes are needed to store the word 'Fermanagh'? [3]

Answers: 1. Data is raw facts and figures with no meaning / context (1); suitable example, e.g. 18, 22, 19 (1); information is data that has been processed / given context so it has meaning (1); suitable example showing the context, e.g. 'Belfast maximum temperature on Monday was 18°C' (1). 2. bit, nibble, byte, kilobyte, megabyte, gigabyte, terabyte. 2 marks for all in correct order; 1 mark if only one item is out of place. 3. (a) 4 (1) (b) 1024 (1) (c) 9 bytes – 9 characters, 1 byte each (1).